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Title

PAYLOAD-AWARE TWO-TIER UAV ROUTE PLANNING FOR LAST-MILE DELIVERY

REVIEW REPORT – I

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Academic Year: 2026–2027

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Abstract

UAV systems for medical and emergency last-mile delivery must balance objectives that conflict: shorter routes carry higher flight risk, low-risk corridors drain more battery, and energy-efficient paths are rarely the fastest. No single fixed trade-off serves all payload types, yet existing planners apply the same objective weights to every mission regardless of what is being carried.

Existing multi-objective planners, including entropy-based, diversity-driven, and crowding-distance Pareto selection methods, encode objective weights as a fixed scalar before path search begins. Payload properties such as fragility class, temperature sensitivity, and delivery urgency have no influence over which Pareto-optimal route is chosen. When the payload changes, operators must re-tune weights manually and re-run the planner, which is not workable under field conditions.

This work proposes a Payload-Aware Two-Tier UAV Route Planning system that separates Pareto front construction from route selection, introducing an explicit algebraic bridge between Tier 1 routing objectives and Tier 2 payload metadata.

Tier 1 runs a multi-objective A*/Dijkstra graph-search over a delivery airspace graph sourced from OpenStreetMap, producing a Pareto-optimal route set across distance, energy consumption, and flight risk. Tier 2 reads four operator-assigned payload tags and derives objective weights through a fixed formula: fragility class scales w_{risk}, delivery urgency scales w_{distance}, and temperature sensitivity scales w_{energy}. These weights score every Pareto point through a utility function; the argmax is selected as the mission route. The system will be evaluated on programmatically generated airspace graphs at three scales (50, 200, and 500 nodes) against single-objective A* and MOPSO baselines, using hypervolume, Pareto spread, and Tier 2 selection accuracy as metrics. Hardware validation will use a Pixhawk-equipped quadcopter through ArduPilot SITL simulation and outdoor flight tests.

Planned experiments will quantify Tier 1 improvements in hypervolume and Pareto spread over both baselines and will determine whether metadata-derived utility scoring selects non-extreme Pareto points for mixed-priority payloads that neither A* nor MOPSO can produce.

The principal contribution is a formula-driven mapping from payload metadata to routing objective weights, connecting mission context directly to Pareto point selection without expert intervention. This replaces context-blind diversity criteria with a direct algebraic link from payload description to route choice, and the Pareto front is computed once and reused across all mission types without re-running the planner.

1. Introduction

Unmanned aerial vehicles (UAVs) have emerged as a compelling platform for last-mile delivery in scenarios where ground transportation is impractical—medical supply chains to remote clinics, emergency drug drops during disaster response, and fragile diagnostic sample transport where road vibration risks contamination. In each scenario the UAV must fly a route from a dispatch hub to a delivery point across an airspace that contains obstacles, regulated no-fly zones, and variable wind corridors. A good route balances three conflicting physical objectives simultaneously: minimising total flight distance, minimising battery energy expenditure, and avoiding high-risk segments.

The standard engineering response to such a conflict is to form a weighted sum of the three objectives into a single scalar cost function and then run a conventional path planner such as A* on it. This scalar collapses the multi-objective problem into a single-objective one, and the operator adjusts the three weights to express the mission's priority before each flight. The approach works for a fixed mission type, but it fails immediately when the payload changes: emergency medicine requires a completely different weight profile from fragile biological samples or disaster-relief bulk supplies. Manually re-tuning and re-running the planner between missions is operationally impractical in time-critical field conditions.

1.1 Background

Multi-objective UAV path planning has attracted sustained research attention because no single route can simultaneously minimise distance, energy, and flight risk—these objectives are structurally in conflict. The dominant approach in the literature is to generate a Pareto-optimal front of non-dominated routes, each representing a different trade-off across the objective space, and then select one route from that front as the mission route.

The critical question that determines the practical value of any such system is: *how is the final route selected from the Pareto front?* Of the three most directly relevant papers reviewed below, one selects the final route with a **front-diversity criterion** (crowding distance) directly, one uses diversity (entropy) only to keep the intermediate archive well-spread and still falls back to a manually weighted scalar sum—re-tuned per mission by a human operator—for the final pick, and the third reports diversity-preserving strategies during front *construction* but stops short of proposing any selection mechanism at all. In every case, delivery context plays no role: none of the three papers allow what the UAV is carrying to influence which route

is chosen, or even address the question of which route should be chosen for a given payload. This is the structural research gap this project addresses.

Autonomous UAV systems for medical and emergency last-mile delivery operate under a structural constraint that distinguishes them from general routing problems: the optimal route trade-off is not fixed—it is a function of what is being carried. A fragile diagnostic sample demands a low-vibration, low-turbulence corridor even at the cost of a much longer flight time. Emergency insulin requires the shortest path regardless of risk. Bulk disaster-relief supplies demand the most energy-efficient route to maximise delivery range.

Payload-awareness itself is not absent from the UAV literature: variable payload mass has been used to adapt flight speed and energy models during trajectory planning [1]. What is absent is any use of payload *metadata*—as opposed to raw mass—to drive the selection of a point on a multi-objective Pareto front, which is the gap this project addresses.

1.2 Motivation

The motivation for this work arises from three real operational scenarios that impose fundamentally incompatible routing requirements:

Fragile medical samples (tissue biopsies, blood samples, pathology specimens) are highly sensitive to vibration, g-forces, and thermal exposure. The dominant routing priority is minimising flight risk—defined here as exposure to turbulent corridors, sudden altitude changes, and wind-exposed segments. A route that adds 40% extra distance but halves the risk of vibration damage is operationally preferable.

Emergency medicine (insulin, epinephrine, antivenom) has a strict time window. The dominant routing priority is minimising route distance to minimise transit time. Flight risk and energy are secondary. For a payload with a 15-minute delivery window, the shortest path wins unconditionally.

Disaster-relief bulk supplies (water purification tablets, bandages, ration packs) weigh significantly more than medical payloads and are not time-critical or fragile. The dominant routing priority is energy efficiency—minimising battery drain to maximise the number of deliveries possible on a single charge cycle.

No existing UAV planner can switch between these three routing strategies based on a payload tag. The proposed two-tier system accomplishes this without any expert intervention or replanning. Payload metadata—fragility class, temperature sensitivity, delivery urgency—has never been used as the sole input to autonomous Pareto point selection in the UAV delivery literature, leaving a structural gap this

project directly addresses.

1.3 Objectives

1. To formulate multi-objective UAV route planning as a structured **Pareto front pre-computation** problem over a delivery airspace graph, theoretically distinguishing the deterministic graph-search approach from stochastic population-based alternatives.

2. To design a **payload metadata schema** and a principled rule-based mapping that translates mission context (fragility class, delivery urgency, temperature sensitivity) into Pareto point selection objective weights without any expert weight tuning.

3. To benchmark Tier 1 route quality and Pareto front coverage against single-objective A* and MOPSO baselines on identical delivery scenarios, measuring hypervolume, Pareto spread, and Inverted Generational Distance (IGD) as three distinct standard multi-objective quality indicators.

4. To quantify **Tier 2 selection accuracy** by measuring alignment between selected route characteristics and payload tag requirements across diverse mission types (fragile medical sample, emergency medicine, disaster-relief supplies).

5. To establish the system's **novelty**: payload metadata is used as the sole input to Pareto point selection, replacing context-blind entropy and diversity-based criteria with a deterministic algebraic mapping from payload description to route choice.

1.4 Scope

This project covers two phases. Phase I delivers a complete, validated software simulation system including the multi-objective A* graph-search engine (Tier 1), the payload-driven route selector (Tier 2), a MOPSO baseline, and a benchmarking suite evaluated on programmatically generated airspace graphs at three scales (50, 200, and 500 nodes). Phase II extends the validated software to physical hardware: a Pixhawk-equipped F450 quadcopter with a Raspberry Pi 5 companion computer, validated through ArduPilot SITL simulation and outdoor flight tests. The project does not address multi-UAV coordination, dynamic obstacle avoidance during flight, or real-time path replanning.

2. Related Work

Multi-objective UAV path planning is a well-studied intersection of graph-search and evolutionary optimisation [2]. Within it, route selection from a computed Pareto front is most commonly guided by either the crowding-distance mechanism introduced by NSGA-II [3], or by an entropy criterion used to keep the archive diverse while a manually weighted scalar sum performs the final pick—in neither case with any regard for delivery context; where diversity is instead pursued only during front construction, no selection mechanism is offered at all. Three papers that are most directly relevant to the proposed work are reviewed below. Table 2.1 provides a structured comparison.

Table 2.1: Summary of related work

S.No.	Author/Year	Proposed Method	Key Findings	Limitations
1	Chokkalingam & Ganesan (2026) [4]	Pareto-optimal particle swarm optimisation (POPSO) with dynamic parameter adaptation and Pareto–entropy archive selection to maintain front diversity during optimisation; the final single route is then read off via a weighted-sum scalarisation whose weights are set empirically per mission.	Entropy-guided archiving improves front diversity and adaptability to real-time environmental changes over static-weight baselines.	Final route choice still reduces to a manually re-tuned weighted sum per mission; payload metadata never incorporated, so fragile samples and emergency medicine receive the same hand-tuned selection procedure from the same front.

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Table 2.1 (continued)

S.No.	Author/Year	Proposed Method	Key Findings	Limitations
2	Zhang et al. (2026) [5]	MEMOBWO: a multi-strategy enhanced multi-objective optimiser that preserves population diversity while *constructing* the Pareto front for UAV path planning in complex urban environments.	Diversity-preserving construction strategies raise hypervolume by 15–31% over baseline algorithms on benchmark and simulated urban scenarios.	Diversity is optimised only during front construction; the paper proposes no mechanism to select a single route from the resulting front, let alone one driven by delivery context or payload type.
3	Ren et al. (2019) [6]	Crowding-distance-based NSGA-II (CDNSGA-II) for multi-objective UAV path planning, balancing distance and safety over an octree-represented urban environment.	CDNSGA-II finds well-spread, collision-free Pareto-optimal paths efficiently in octree-modelled urban airspace.	Crowding distance encodes only the geometric spread of the front; it cannot represent urgency, fragility, or thermal sensitivity, and mission context plays no role in selection.

2.1 Research Gap

All three reviewed papers treat Pareto-front diversity as the central design concern in multi-objective UAV path planning, but only two of them use it to *select* a route: the crowding distance of [6] picks a single point directly from the front, while [4] uses entropy only to keep the archive *diverse* during optimisation and falls back to a manually weighted scalar sum—re-tuned per mission by a human operator—to make the actual final pick. The diversity-preserving strategies of [5] operate one stage earlier, during front *construction*, and the paper proposes no selection mechanism at

all. Whichever stage diversity is applied at, none of the three let delivery context (what payload is being carried) influence, or replace the need for expert re-tuning of, the route that is finally selected. This is the structural gap the proposed system directly addresses.

The gap is not a minor implementation detail—it is a fundamental design assumption. Entropy and crowding distance are both defined purely over the geometry of the Pareto front; population-diversity strategies during construction are defined purely over the search process itself. None of these mechanisms has an interface through which external mission context can enter. Extending them to incorporate payload information would require replacing the selection criterion entirely, which is precisely what the proposed Tier 2 utility scoring mechanism does.

Three specific gaps addressed by this work are as follows. **Gap 1 (No payload-aware Pareto selection):** Existing UAV multi-objective planners either apply a fixed geometric selection criterion independent of payload type, or offer no selection mechanism at all. The proposed Tier 2 introduces a formula-driven mapping from four payload tags directly to Pareto point selection. **Gap 2 (No re-use across mission types):** Existing planners re-run the full multi-objective search when the mission context changes. The proposed two-tier separation pre-computes the Pareto front once (Tier 1) and allows instantaneous route selection for any new payload tag (Tier 2), without re-running the planner. **Gap 3 (No algebraic bridge between delivery context and routing objectives):** The mapping between a payload description and the routing trade-off it implies (e.g., fragility $\rightarrow$ minimise risk, urgency $\rightarrow$ minimise distance) has never been formalised algebraically. The proposed weight-derivation formula provides this bridge, making the mapping deterministic, transparent, and reproducible.

3. Problem Statement

Existing UAV route planners encode distance, energy consumption, and flight risk into a **single hand-tuned weighted-sum scalar** before path search begins, permanently fixing the objective trade-off regardless of what is being carried. When the payload context changes—from fragile diagnostic samples requiring vibration-minimal routing to time-critical emergency medicine requiring shortest-path routing—an expert must manually re-tune the planner weights and re-run the search from scratch. This intervention is **infeasible in field conditions**.

Formally, given: (i) an airspace graph $G = (V, E)$ with three-objective edge cost vectors $\mathbf{c}(e) = [d(e),\, \varepsilon(e),\, \rho(e)]$ representing geodesic distance, estimated energy consumption, and flight risk respectively; (ii) a Pareto-optimal front $\mathcal{P}$ of non-dominated routes pre-computed over G; and (iii) a payload metadata tuple $(F, T, U, P_{\text{lvl}})$ encoding fragility class $F \in \{1 \dots 5\}$, temperature sensitivity $T \in \{1 \dots 3\}$, delivery urgency $U \in \{1 \dots 3\}$, and priority level $P_{\text{lvl}} \in \{1 \dots 5\}$—the problem is to select the route $p^* \in \mathcal{P}$ that best satisfies the routing requirements implied by the payload metadata, without expert re-tuning and without re-running the multi-objective search.

The inputs are measurable quantities the operator can assign from domain knowledge (e.g., "this vial is fragile: $F = 5$; delivery is non-urgent: $U = 1$"). The constraints are that selection must be deterministic, reproducible, and complete within one second of payload tag entry. The success criterion is that the selected route characteristics (distance, energy, risk) align with the payload's stated requirements as verified by a Tier 2 selection accuracy metric evaluated across multiple payload presets.

4. Justification for the Problem Formulation

4.1 Need for the Proposed Work

The gap established in Chapter 2 is not merely a modelling omission—it has a direct operational consequence. Manually re-tuning three objective weights and re-running a multi-objective search between every dispatch requires an expert on hand at the point of launch, which is unavailable in a time-critical emergency deployment. Any system that requires this intervention is, in practice, unusable in the field regardless of its routing quality on paper.

This is precisely what motivates separating Pareto front construction from payload-driven selection: the front is computed once, offline, for a delivery region, and every subsequent mission needs only a constant-time metadata lookup—no expert, no re-planning, no delay between payload tagging and mission file generation.

The proposed system resolves this by separating Pareto front generation from mission-context-driven route selection. The Pareto front is computed once for a delivery region and stored. For each new mission, only Tier 2 runs—reading the payload tag and selecting the appropriate route from the pre-computed front in near-instantaneous time, with no re-planning and no expert weight adjustment required.

4.2 Feasibility

Technical feasibility: The Tier 1 multi-objective A*/Dijkstra algorithm is a well-studied graph-search extension. The Tier 2 utility scoring is a closed-form algebraic formula executable in microseconds on any hardware. Both components are implementable in Python with standard open-source libraries (NetworkX [7], NumPy). Hardware deployment on a Raspberry Pi 5 companion computer is feasible given the computational simplicity of Tier 2 route selection (< 1 s end-to-end from payload tag input to mission file output).

Operational feasibility: The system requires only four integer tag inputs from the operator, eliminating the need for weight-tuning expertise. The ArduPilot SITL simulation environment allows complete end-to-end validation before outdoor flights, reducing hardware risk to an acceptable level.

Economic feasibility: All software components use open-source libraries and protocols (osmnx, NetworkX, NumPy, matplotlib, MAVLink [8]). The hardware platform—F450 quadcopter frame, Pixhawk flight controller, Raspberry Pi 5—uses standard, commercially available components at modest cost. No specialised or

proprietary software licences are required.

Schedule feasibility: Phase I (software simulation and benchmarking) is achievable within the first semester aligned to Review I. Phase II (hardware integration and outdoor testing) is planned for the second semester, with milestones aligned to the Review II and Final Review schedule as shown in Chapter 6.

4.3 Expected Contribution

The contribution here is not the multi-objective search itself—Pareto-optimal graph-search is an established technique—but the fact that the bridge connecting Tier 1's routing objectives to Tier 2's payload metadata is algebraic rather than a lookup table or an expert-set rule. Because every weight is a function of all three raw scores simultaneously (Chapter 5), the resulting utility function can select any point on the front, not only its extremes, and does so without a human deciding which extreme applies to which payload. No prior UAV delivery system grants payload metadata this role: existing planners treat route selection as a front-diversity problem, blind to what is being carried. Practically, this means one pre-computed front serves every mission profile a delivery region will ever see, with route choice reduced to a sub-second metadata lookup instead of a re-planning exercise.

5. Methodology

The proposed system is built on a single architectural decision: **separate Pareto front construction (Tier 1) from route selection (Tier 2)**. This separation eliminates the need to re-run a multi-objective search each time the payload changes. The Pareto front is computed once for a delivery region and stored persistently. For each new mission, only Tier 2 runs—reading the payload tag and selecting the appropriate route from the pre-computed front in near-instantaneous time.

5.1 Architectural Design for Proposed System

Figure 5.1 shows the complete two-tier system architecture. Airspace graph data from OpenStreetMap enters Tier 1, which performs a multi-objective A*/Dijkstra search and stores the resulting Pareto-optimal route set. For each new mission, Tier 2 reads the four operator-assigned payload tags, derives objective weights algebraically, scores every stored Pareto point via a utility function, and outputs the arg max as the selected mission route along with an ArduPilot-compatible `.waypoints` file.

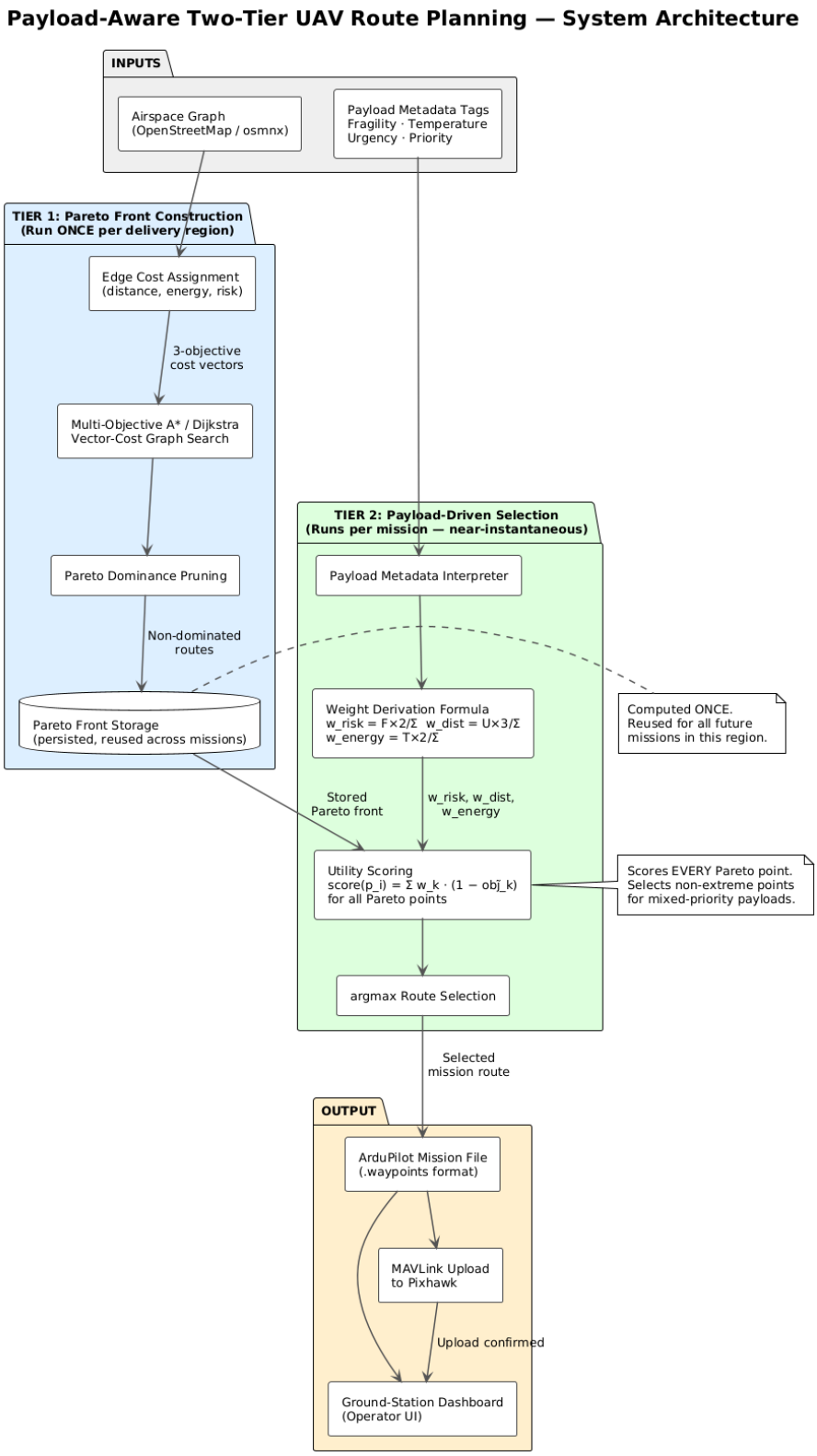


Figure 5.1: Overall system architecture: Two-Tier Payload-Aware UAV Route Planner. Tier 1 pre-computes the Pareto front once per delivery region; Tier 2 selects the optimal route for each new payload tag using the stored front.

The airspace graph $G = (V, E)$ is extracted from OpenStreetMap using the

`osmnx` Python library [9]. Each edge $e \in E$ carries a cost vector:

$$\mathbf{c}(e) = \Big[d(e),\ \varepsilon(e),\ \rho(e)\Big]$$

where $d(e)$ is the geodesic distance of the flight segment (km), $\varepsilon(e)$ is the estimated energy consumption (proxy: $d(e)\times$ altitude-change factor), and $\rho(e) \in [0, 1]$ is the flight risk score derived from obstacle density, proximity to no-fly zones, and wind exposure (simulated programmatically for reproducible benchmarking).

The three optimisation objectives are: (1) minimise route distance d; (2) minimise energy consumption ε; and (3) minimise flight risk ρ. A route p dominates route q if p is at least as good as q on all three objectives and strictly better on at least one. The Pareto front $\mathcal{P}$ is the set of all non-dominated routes.

Tier 2 Algebraic Bridge. Each of the four payload metadata tags maps to exactly one Tier 1 routing objective as shown in Table 5.1.

Table 5.1: Semantic mapping: payload metadata tags to Tier 1 routing objectives

Payload Tag	**Range**	**Maps to Tier 1 Objective**	**Reasoning**
Fragility class F	1–5	w_{risk} (flight risk weight)	Fragile items need low-vibration, low-risk corridors
Temperature sensitivity T	1–3	w_{energy} (energy weight)	Temp-sensitive items need short exposure time $\Rightarrow$ fast flight
Delivery urgency U	1–3	w_{dist} (distance weight)	Time-critical deliveries need the shortest path
Priority level P_{lvl}	1–5	Tie-breaker amplifier	Scales the dominant weight when scores are close

From the payload tags, three routing objective weights are derived algorithmically with no expert input:

$$r_{\text{raw}} = F \times 2, \quad d_{\text{raw}} = U \times 3, \quad e_{\text{raw}} = T \times 2 \tag{5.1}$$

$$w_{\text{risk}} = \frac{r_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}}, \quad w_{\text{dist}} = \frac{d_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}}, \quad w_{\text{energy}} = \frac{e_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}} \tag{5.2}$$

The multipliers (2 for fragility and temperature, 3 for urgency) reflect the domain insight that urgency has a higher penalty sensitivity than fragility or thermal constraints in the UAV medical delivery context.

For each Pareto point p_i with objective values $(d_i, \varepsilon_i, \rho_i)$, min-max normalised values $\tilde{d}_i$, $\tilde{\varepsilon}_i$, $\tilde{\rho}_i$ are computed over the full Pareto front $\mathcal{P}$. The utility score is:

$$\text{score}(p_i) = w_{\text{dist}} \cdot (1 - \tilde{d}_i) + w_{\text{energy}} \cdot (1 - \tilde{\varepsilon}_i) + w_{\text{risk}} \cdot (1 - \tilde{\rho}_i) \tag{5.3}$$

The selected mission route is $p^* = \arg\max_{p_i \in \mathcal{P}} \text{score}(p_i)$. For high-priority missions ($P_{\text{lvl}} \geq 4$), the dominant weight is amplified by $0.1 \times w_{\max}$ to sharpen the selection.

The Tier 1 and Tier 2 activity flows are shown in Figures 5.2 and 5.3 respectively.

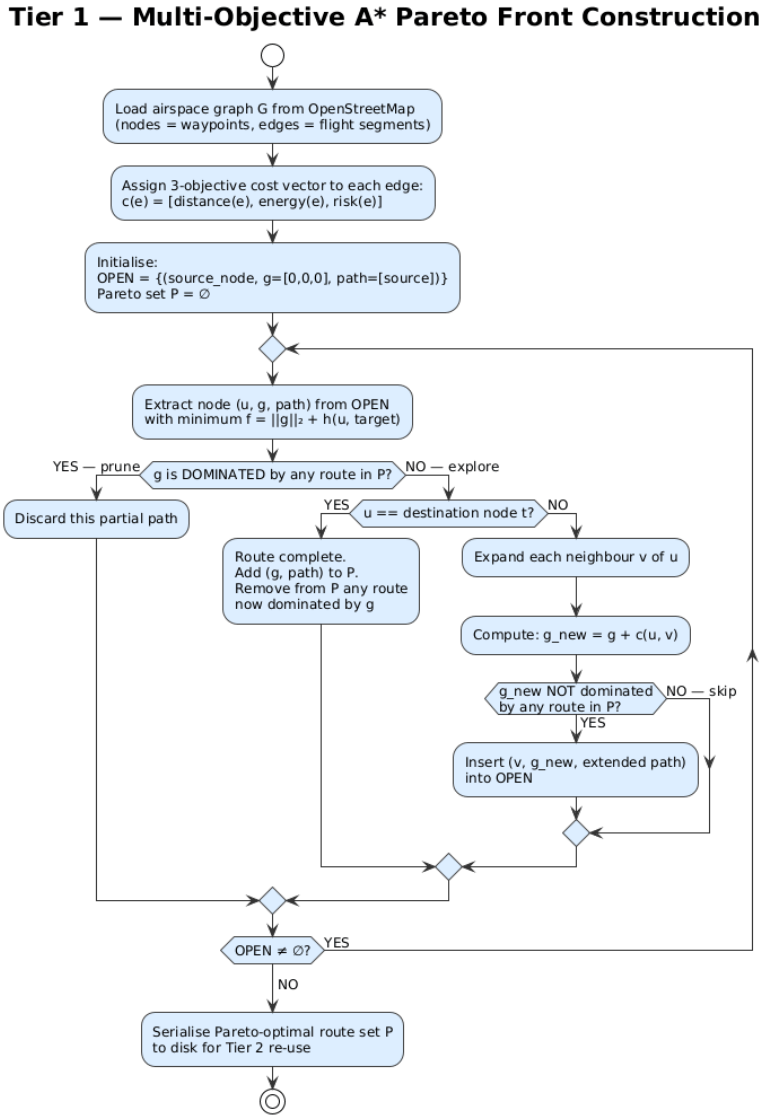


Figure 5.2: Tier 1 activity flow: Multi-Objective A* Pareto Front Construction. The open set is explored in order of estimated total cost until no further non-dominated routes exist.

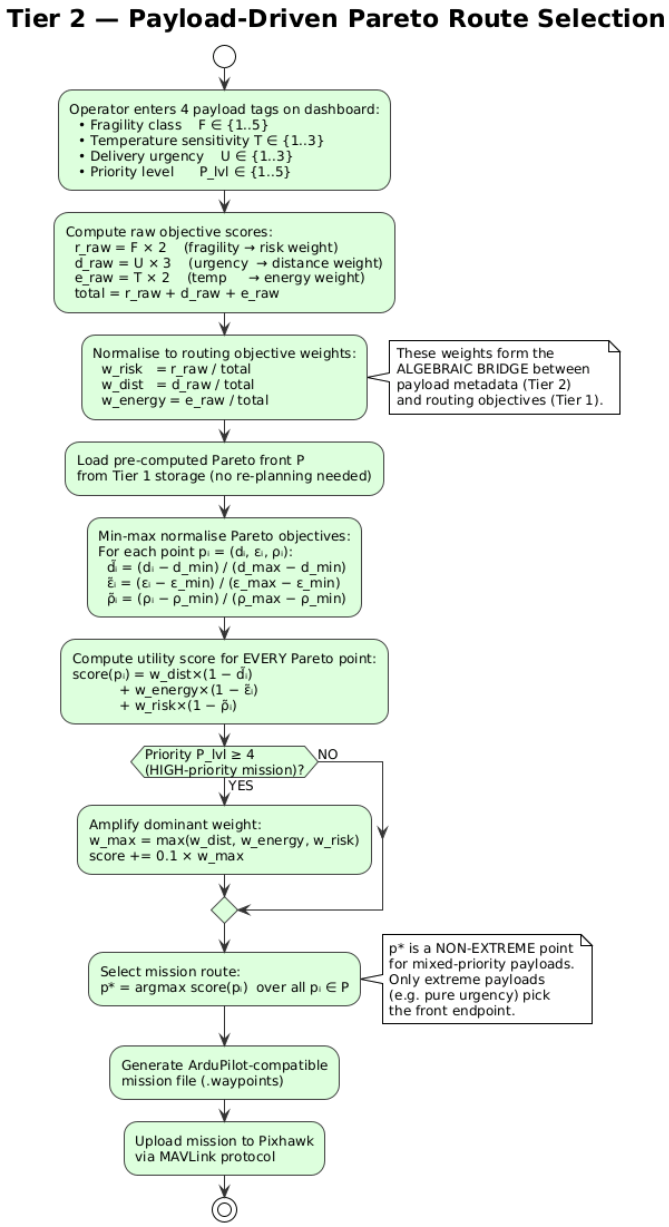


Figure 5.3: Tier 2 activity flow: Payload-Driven Pareto Selection. Payload tags are algebraically mapped to weights, which score every Pareto point via a utility function; the arg max is selected as the mission route.

5.2 Modules

5.2.1 Module 1: Airspace Graph Builder

Extracts the road/path network from OpenStreetMap for a specified delivery region using `osmnx`. Assigns three objective costs to each edge: geodesic distance, energy (distance × altitude-change factor), and flight risk (simulated threat zones and obstacle density scored 0–1 using a configurable hazard map). Output: a directed weighted

graph G stored as a serialised NetworkX object ready for Tier 1 search.

5.2.2 Module 2: Multi-Objective A*/Dijkstra Engine (Tier 1)

Implements the multi-objective A* algorithm maintaining a cost vector $\mathbf{g}$ (not a scalar) for each partial path. Paths dominated by already-discovered routes are pruned early. The `UpdateParetoFront` subroutine adds a new route and removes any previously stored routes it dominates. Output: a deterministic, reproducible Pareto front $\mathcal{P}$ serialised to disk for Tier 2 re-use across all future missions in the same delivery region.

5.2.3 Module 3: Payload Metadata Interpreter (Tier 2)

Implements the payload-driven route selection algorithm: reads the four payload tags, derives objective weights via Equations (5.1)–(5.2), applies min-max normalisation to Pareto objectives, computes a utility score for every Pareto point via Equation (5.3), and returns the $\arg\max$ route. Optionally amplifies the dominant weight for high-priority missions ($P_{\text{lvl}} \geq 4$). Generates an ArduPilot-compatible `.waypoints` mission file for upload.

5.2.4 Module 4: Testing and Evaluation

Evaluates all methods on three scenario scales: S1 ($\approx$50 nodes, low obstacle density), S2 ($\approx$200 nodes, medium density), and S3 ($\approx$500 nodes, high density). The MOPSO baseline [10] runs with swarm size 100, 200 iterations, and a Pareto archive of 100 solutions, repeated 30 times per scenario to obtain mean $\pm$ standard deviation. Metrics include Hypervolume (HV), Pareto spread (the standard Δ metric, capturing extent and uniformity of the front), Inverted Generational Distance (IGD) computed against a reference front formed from the union of all non-dominated solutions found across every method and repetition for each scenario, and Tier 2 selection accuracy across four payload presets. Phase II testing extends to ArduPilot SITL validation and outdoor flight tests on an F450 quadcopter with Pixhawk flight controller, measuring route adherence (GPS position error vs. planned route) and end-to-end latency.

6. Project Timeline Chart

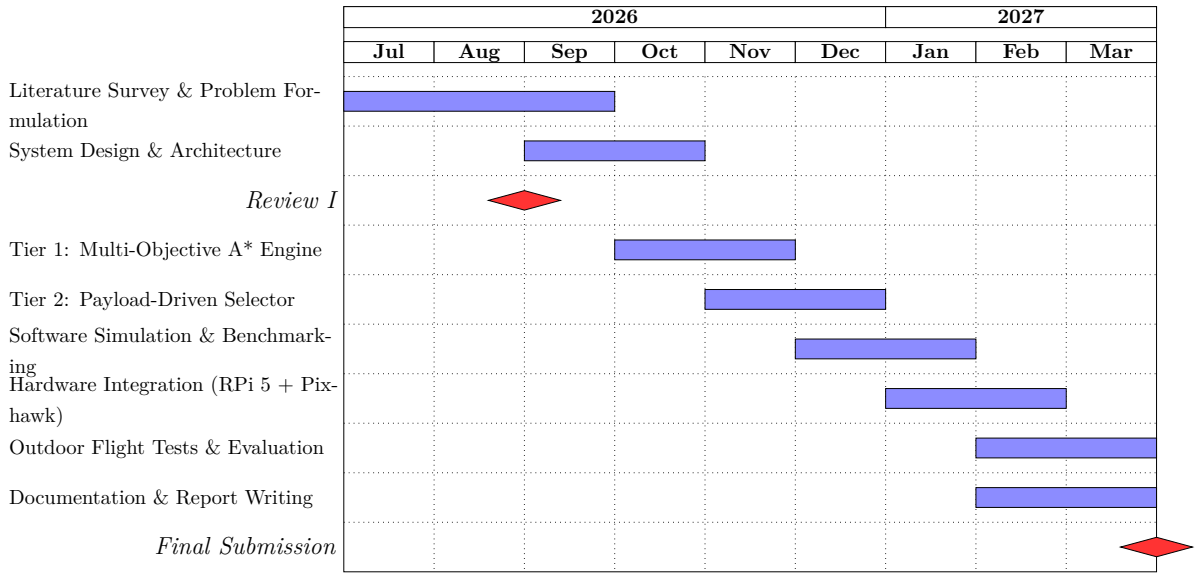


Figure 6.1: Project timeline (July 2026 – March 2027). Red diamonds mark review and submission milestones.

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